

Economics: A Unified Research Reference

Foundations, Models, and Empirical Practice in Microeconomics and Macroeconomics

Study and Research Compendium

Abstract

This reference synthesizes core microeconomic and macroeconomic theory into a single working document intended for study and applied research. It presents unifying concepts, canonical models, and their governing equations, followed by an integrated formula sheet, glossary, and reference list. Emphasis is placed on the conditions under which theoretical predictions hold and the empirical challenges — measurement error, omitted variables, selection, simultaneity, and external validity — that arise when those predictions are tested against data.

Keywords: microeconomics, macroeconomics, equilibrium, elasticity, growth, stabilization policy, applied econometrics

Table 1. Unifying concepts across microeconomics and macroeconomics.

Concept	Central question	Key mechanism
Scarcity	What must be sacrificed?	Opportunity cost and constrained choice
Marginal analysis	Should one more unit be chosen?	Compare marginal benefit and marginal cost
Elasticity	How responsive is behavior?	Percentage change relative to percentage change
Equilibrium	When are plans consistent?	Prices, quantities, expectations, and incentives align
Efficiency	Are gains from trade exhausted?	Marginal willingness to pay equals marginal cost
Market failure	Why might markets misallocate?	Externalities, information, market power, public goods
Macroeconomic stabilization	How are recessions and inflation managed?	Fiscal, monetary, and financial policy
Growth	Why do living standards rise?	Capital, human capital, institutions, technology

A note on mathematical notation: equations are compact and use standard ASCII notation so the PDF remains portable. Variables are defined near first use. The document is designed as a study and research reference, not as legal, investment, or policy advice.

1. Scarcity, Choice, and Opportunity Cost

Constrained optimization

Every economic decision is a choice among alternatives under a binding resource constraint. The opportunity cost of a choice is the value of the next-best alternative forgone, not merely its stated price.

Production possibilities

The production possibilities frontier illustrates the maximum combinations of two goods an economy can produce given fixed resources and technology, and its bowed-out shape reflects increasing opportunity cost.

Key relationships:

Opportunity cost = value of next-best alternative forgone

PPF slope = $-(MC \text{ of good } X / MC \text{ of good } Y)$

Empirically, scarcity, choice, and opportunity cost should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

2. Supply, Demand, and Market Equilibrium

Market clearing

Equilibrium price and quantity occur where the supply and demand schedules intersect, so that quantity supplied equals quantity demanded and no participant has an incentive to unilaterally change behavior.

Elasticity

Price elasticity of demand measures the percentage change in quantity demanded for a given percentage change in price, governing how much of a tax burden falls on consumers versus producers.

Key relationships:

$$Q_d(P) = Q_s(P) \text{ at equilibrium}$$

$$E_d = (dQ/dP)(P/Q)$$

Empirically, supply, demand, and market equilibrium should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

3. Consumer Theory

Utility maximization

A rational consumer allocates a fixed budget across goods so that the marginal utility per dollar spent is equalized across all goods purchased.

Income and substitution effects

A price change alters consumption through a substitution effect, holding utility constant, and an income effect arising from the change in purchasing power.

Key relationships:

$$MU_x/P_x = MU_y/P_y$$

$$\text{Total effect} = \text{substitution effect} + \text{income effect}$$

Empirically, consumer theory should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

4. Firm Behavior and Market Structure

Profit maximization

A firm maximizes profit by producing the quantity at which marginal revenue equals marginal cost, regardless of market structure.

Competition versus monopoly

Under perfect competition, free entry and exit drive price to marginal cost and long-run economic profit to zero; a monopolist restricts output to raise price above marginal cost, creating deadweight loss.

Key relationships:

$$MR = MC$$

$$\text{Competitive long run: } P = MC = \min ATC$$

Empirically, firm behavior and market structure should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

5. Market Failure and Public Policy

Externalities

When private costs or benefits diverge from social costs or benefits, unregulated markets over- or under-produce relative to the efficient quantity.

Information and public goods

Asymmetric information can cause adverse selection or moral hazard, while non-excludable, non-rival goods are underprovided without collective action.

Key relationships:

$$MSC = MSB \text{ at the efficient quantity}$$

$$\text{Deadweight loss} = 0.5 \times |\Delta Q| \times |\Delta P| \text{ (linear case)}$$

Empirically, market failure and public policy should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

Microeconomic extension: Behavioral deviations

Observed choices sometimes depart from strict expected-utility predictions, motivating behavioral extensions such as loss aversion and present bias.

Microeconomic extension: Game-theoretic strategy

Strategic interaction between a small number of firms is best analyzed with Nash equilibrium concepts rather than price-taking assumptions.

Microeconomic extension: Price discrimination

Firms with market power can raise profit by charging different prices to segments with different willingness to pay, subject to arbitrage constraints.

Microeconomic extension: Labor supply

Wage changes generate offsetting income and substitution effects on hours worked, so aggregate labor supply elasticity is an empirical question.

Microeconomic extension: Capital markets

Interest rates equilibrate the supply of loanable funds from savers with the demand for funds from borrowers and investors.

Microeconomic extension: Trade and comparative advantage

Countries gain from specializing according to relative, not absolute, productivity differences, then trading for goods produced by trading partners.

Microeconomic extension: Auctions and mechanism design

The rules of an auction shape bidding incentives and revenue; well-designed mechanisms can approximate efficient allocation under private information.

Microeconomic extension: Risk and insurance

Risk-averse agents value insurance above its actuarially fair price, creating gains from trade that are limited by adverse selection and moral hazard.

Microeconomic extension: Human capital

Education and training raise future earnings, so schooling choices resemble an investment decision with returns realized over a working lifetime.

Microeconomic extension: Environmental economics

Pigouvian taxes and cap-and-trade systems aim to internalize externalities by aligning private incentives with social cost.

6. National Income Accounting

Expenditure approach

Gross domestic product is measured as the sum of consumption, investment, government purchases, and net exports over a given period.

Saving-investment identity

In an open economy, national saving need not equal domestic investment; the gap is financed through net exports and international capital flows.

Key relationships:

$$Y = C + I + G + NX$$

$$S = I + NX$$

Empirically, national income accounting should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

7. Money, Prices, and Inflation

Quantity theory

In the long run, the price level moves proportionally with the money supply, given stable velocity and real output growth.

Fisher relationship

Nominal interest rates adjust to reflect both the real return lenders require and the inflation they expect over the loan horizon.

Key relationships:

$$MV = PY$$
$$i \approx r + \pi^e$$

Empirically, money, prices, and inflation should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

8. Short-Run Fluctuations

IS-LM framework

Goods-market equilibrium and money-market equilibrium jointly pin down output and the interest rate when prices are sticky in the short run.

Phillips curve

Unemployment below its natural rate tends to push inflation above expectations, subject to supply shocks that shift the trade-off.

Key relationships:

$$\pi = \pi^e - \beta(u - u^*) + v$$
$$\Delta k = sf(k) - (\delta + n)k$$

Empirically, short-run fluctuations should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

9. Fiscal and Monetary Policy

Stabilization tools

Governments and central banks use spending, taxation, and interest rate or balance-sheet policy to smooth business cycles, subject to lags and credibility constraints.

Debt sustainability

The evolution of the debt-to-GDP ratio depends on the gap between the interest rate and the growth rate, and on the primary fiscal balance.

Key relationships:

$$\Delta b \approx (r - g)b - ps$$

Automatic stabilizers offset cycles without new legislation

Empirically, fiscal and monetary policy should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

10. Open Economy Macroeconomics

Exchange rates

The real exchange rate adjusts the nominal rate for relative price levels, determining the relative cost of domestic and foreign goods.

Policy trilemma

A country cannot simultaneously maintain free capital flows, a fixed exchange rate, and independent monetary policy.

Key relationships:

$$q = e P^* / P$$

Trilemma: pick at most two of free capital flow, fixed FX, monetary autonomy

Empirically, open economy macroeconomics should be studied with attention to measurement error, omitted variables, selection, simultaneity, and external validity. A clean table or equation is only the beginning. The researcher must ask who changes behavior, over what horizon, with what information, under which incentives, and with what distributional consequences. Policy conclusions are strongest when theory, data, institutional detail, and robustness checks point in the same direction.

Applied extension: Business cycle indicators

Leading, coincident, and lagging indicators are combined to date turning points in aggregate economic activity.

Applied extension: Central bank credibility

Policy is more effective when private expectations align with announced targets, reducing the output cost of disinflation.

Applied extension: Financial intermediation

Banks transform short-term liabilities into long-term assets, creating maturity mismatch risk that motivates prudential regulation.

Applied extension: Unemployment types

Frictional, structural, and cyclical unemployment have distinct causes and call for different policy responses.

Applied extension: Growth accounting

Output growth is decomposed into contributions from capital deepening, labor input growth, and total factor productivity.

Applied extension: Sovereign debt crises

Rollover risk rises sharply when the interest-growth differential turns unfavorable and market access becomes uncertain.

Applied extension: Exchange rate regimes

Fixed, floating, and managed regimes trade off nominal stability against monetary policy independence.

Applied extension: Income inequality

Distributional outcomes depend on the interaction of factor returns, taxation, and transfer programs over the life cycle.

Applied extension: Development economics

Institutions, human capital, and access to credit help explain persistent differences in living standards across countries.

Applied extension: Applied econometrics

Identification strategies such as instrumental variables and difference-in-differences aim to isolate causal effects from observational data.

11. Comprehensive Microeconomic Concept Atlas

Term	Definition
Marginal utility	The additional satisfaction from consuming one more unit of a good.
Consumer surplus	The gap between what consumers are willing to pay and what they pay.
Producer surplus	The gap between the market price and a producer's marginal cost.
Price ceiling	A legal maximum price that, if binding, causes a shortage.
Price floor	A legal minimum price that, if binding, causes a surplus.
Cross-price elasticity	Responsiveness of demand for one good to a price change in another.
Income elasticity	Responsiveness of demand to a change in consumer income.
Monopolistic competition	Many firms sell differentiated products with some pricing power.
Oligopoly	A market with few firms whose decisions are strategically interdependent.

Term	Definition
Pareto efficiency	An allocation where no one can be made better off without harming another.

12. Comprehensive Macroeconomic Concept Atlas

Term	Definition
Potential output	The output level consistent with normal resource use and stable inflation.
Output gap	The percentage difference between actual and potential output.
Natural rate of unemployment	The unemployment rate consistent with stable inflation.
Real interest rate	The nominal interest rate adjusted for expected inflation.
Current account	The sum of net exports, net income, and net transfers with the rest of the world.
Business cycle	Recurring, non-periodic fluctuations in aggregate economic activity.
Monetary base	Currency in circulation plus bank reserves held at the central bank.
Fiscal multiplier	The change in output per unit change in government spending or taxes.
Sticky prices	Prices that adjust slowly to shocks, allowing short-run demand effects on output.
Balance of payments	The record of a country's transactions with the rest of the world.

13. Appendix: Canonical Model Summaries

Supply and demand

Two schedules relating price to quantity; intersection defines market-clearing price and quantity absent frictions.

Consumer choice

Utility maximization subject to a budget constraint; yields demand curves and comparative statics from price and income changes.

Perfect competition

Price-taking firms produce where marginal cost equals price; free entry drives long-run economic profit to zero.

Monopoly

A single price-searching firm sets marginal revenue equal to marginal cost, producing less and charging more than the competitive benchmark.

IS-LM

Goods-market and money-market equilibrium loci jointly determine output and the interest rate in the short run under sticky prices.

Solow growth model

Capital accumulation, depreciation, and population growth determine a steady state; technology shifts the steady state itself.

Phillips curve

A short-run trade-off between inflation and unemployment, shifted by inflation expectations and supply shocks.

Mundell-Fleming

Extends IS-LM to an open economy, showing how capital mobility and the exchange-rate regime shape policy effectiveness.

14. Integrated Formula Sheet and Glossary

Area	Formula or condition	Interpretation
Choice	$MB = MC$	Optimal activity level in interior solutions
Consumer	$MU_x/P_x = MU_y/P_y$	Equal marginal utility per dollar across goods
Demand	$E_d = (dQ/dP)(P/Q)$	Price elasticity of demand
Firm	$MR = MC$	Profit-maximizing output for price searcher
Competition	$P = MC = \min ATC$	Long-run competitive equilibrium
Externality	$MSC = MSB$	Efficient social quantity
GDP	$Y = C + I + G + NX$	Expenditure identity
Saving	$S = I + NX$	Open-economy saving-investment relation
Money	$MV = PY$	Quantity equation
Fisher	$i \approx r + \pi^e$	Nominal rate equals real rate plus expected inflation
Phillips	$\pi = \pi^e - \beta(u - u^*) + v$	Inflation, expectations, slack, shocks
Solow	$\Delta k = sf(k) - (\delta + n)k$	Capital accumulation per worker
Debt	$\Delta b \approx (r - g)b - ps$	Debt ratio sustainability
Exchange	$q = eP^*/P$	Real exchange rate
Regression	$y = \beta_0 + \beta_1 x + u$	Empirical conditional relationship

Adverse selection. Hidden characteristics before exchange make participation non-random.

Allocative efficiency. Resources flow to uses where marginal benefit equals marginal cost.

Automatic stabilizer. Budget component that offsets cycles without new legislation.

Comparative advantage. Lower opportunity cost in producing a good.

Deadweight loss. Lost total surplus from distorted quantities.

Fiscal dominance. Monetary policy constrained by government financing needs.

Liquidity trap. Very low rates make money and bonds close substitutes, weakening conventional policy.

Moral hazard. Hidden action after exchange changes risk-taking or effort.

Natural monopoly. Lowest-cost provision occurs with one firm over relevant demand.

Potential output. Sustainable output consistent with normal resource use and stable inflation.

Ricardian equivalence. Tax cuts do not raise demand if households fully anticipate future taxes.

Stagflation. High inflation combined with weak output or high unemployment.

15. References

- Acemoglu, D., Laibson, D., and List, J. Economics. Pearson.
- Akerlof, G. A. The market for lemons: quality uncertainty and the market mechanism. *Quarterly Journal of Economics*.
- Blanchard, O. Macroeconomics. Pearson.
- Cowen, T. and Tabarrok, A. Modern Principles of Economics. Worth.
- Dornbusch, R., Fischer, S., and Startz, R. Macroeconomics. McGraw-Hill.
- Friedman, M. The role of monetary policy. *American Economic Review*.
- Krugman, P., Obstfeld, M., and Melitz, M. International Economics: Theory and Policy. Pearson.
- Mankiw, N. G. Principles of Economics. Cengage.
- Mas-Colell, A., Whinston, M., and Green, J. Microeconomic Theory. Oxford University Press.
- Piketty, T. Capital in the Twenty-First Century. Harvard University Press.
- Romer, D. Advanced Macroeconomics. McGraw-Hill.
- Samuelson, P. A. The pure theory of public expenditure. *Review of Economics and Statistics*.
- Stiglitz, J. E. and Walsh, C. E. Economics. W. W. Norton.
- Varian, H. R. Intermediate Microeconomics. W. W. Norton.
- World Bank. World Development Report series.